

Rollno :241001314

IMPLEMENT MINIMAX ALGORITHM

Python code:

```
def minimax(depth, nodeIndex, isMax, scores, h):

    if depth == h:
        return scores[nodeIndex]

    if isMax:
        return max(
            minimax(depth + 1, nodeIndex * 2, False, scores, h),
            minimax(depth + 1, nodeIndex * 2 + 1, False, scores, h)
        )

    else:
        return min(
            minimax(depth + 1, nodeIndex * 2, True, scores, h),
            minimax(depth + 1, nodeIndex * 2 + 1, True, scores, h)
```

)

```
scores = [3, 5, 6, 9, 1, 2, 0, -1]
```

```
h = 3
```

```
print("Optimal value:", minimax(0, 0, True, scores, h))
```

Output:

Optimal value: 5

Result:

Thus the program of using the MINIMAX algorithm to evaluate the optimal move in a game tree was executed successfully.